11 Chemistry
Unit 2 Topic Test 4: Gases II (Gas Stoichiometry)

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water, including calculations for reactions that involve gases. This topic test assesses gas stoichiometry (mass, amount and gas volume at SLC and at other conditions using pV = nRT), molar volume, the molar mass of a gas produced by a reaction, and limiting and excess reactants in gas-producing reactions.

Task Overview

Section A – Multiple Choice: 10 questions, 1 mark each (10 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 4 questions (30 marks). Answer in this booklet.

Conditions

- Test conditions (revision topic test)
- VCE Chemistry Data Book provided
- Permitted materials: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator (no CAS).
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 46 mins writing time (about 1.15 minutes per mark)

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

SECTION A – Multiple-choice questions (10 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

Double-glazed windows are often filled with argon, Ar, to reduce heat loss. One window unit holds 1.50 L of argon, measured at SLC.

The mass of argon in the window unit is closest to

A. 2.67 g

B. 2.41 g

C. 1.09 g

D. 0.0605 g

Question 2

At the same temperature and pressure, 1.0 L of methane, CH_4, and 1.0 L of carbon dioxide, CO_2, contain the same

A. mass of gas.

B. number of atoms.

C. density.

D. number of molecules.

Question 3

The molar volume of a gas at SLC, 24.8 L mol^{-1}, is the volume occupied by

A. 1 mol of any gas at 25 °C and 100 kPa.

B. 1 g of any gas at 25 °C and 100 kPa.

C. 1 mol of any gas at 0 °C and 100 kPa.

D. 1 mol of oxygen gas only, at 25 °C and 100 kPa.

Question 4

Calcium hydride, CaH_2, is used as a portable source of hydrogen gas, H_2. It reacts with water according to the equation

$$CaH_2(s) + 2H_2O(l) \rightarrow Ca(OH)_2(s) + 2H_2(g)$$

The volume of hydrogen produced when 10.5 g of CaH_2 reacts completely, measured at 15.0 °C and 101 kPa, is closest to

A. 5.91 L

B. 0.616 L

C. 11.8 L

D. 12.4 L

Use the following information to answer Questions 5 and 6.

Propane, C_3H_8, is the fuel in many gas barbecues. It burns according to the equation

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$$

Question 5

When 2.0 L of propane burns completely, the volume of oxygen used, measured at the same temperature and pressure, is

A. 10.0 L

B. 2.0 L

C. 6.0 L

D. 0.40 L

Question 6

A mixture of 20 mL of propane and 80 mL of oxygen reacts until the reaction is complete. All volumes are measured at the same temperature and pressure, at which water is a liquid with negligible volume.

The total volume of gas present after the reaction is

A. 48 mL

B. 116 mL

C. 100 mL

D. 52 mL

Question 7

Under which one of the following sets of conditions is the molar volume of a gas largest?

A. 25 °C and 200 kPa

B. 100 °C and 100 kPa

C. 0 °C and 100 kPa

D. 25 °C and 100 kPa

Question 8

Small amounts of nitrogen gas, N_2, can be made in the laboratory by gently heating ammonium nitrite, NH_4NO_2.

$$NH_4NO_2(s) \rightarrow N_2(g) + 2H_2O(g)$$

The mass of NH_4NO_2 needed to produce 1.50 L of N_2, measured at SLC, is closest to

A. 7.74 g

B. 4.29 g

C. 3.87 g

D. 1.69 g

Question 9

Some drain cleaners contain sodium hydroxide, NaOH, and granules of aluminium, Al. In water they react to produce hydrogen gas, H_2, which helps to stir up the blockage.

$$2Al(s) + 2NaOH(aq) + 6H_2O(l) \rightarrow 2NaAl(OH)_4(aq) + 3H_2(g)$$

In a test, 0.270 g of aluminium was added to 15.0 mL of 0.500 M NaOH(aq). Water is in excess.

The volume of hydrogen, measured at SLC, produced when the reaction stops is closest to

A. 0.279 L

B. 0.372 L

C. 0.186 L

D. 0.558 L

Question 10

In a car's catalytic converter, carbon monoxide, CO, reacts with oxygen, O_2, to form carbon dioxide, CO_2.

$$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$$

The volume of CO_2, measured at SLC, formed when 0.140 g of CO reacts completely is closest to

A. 62.0 mL

B. 124 mL

C. 248 mL

D. 3.47 L

END OF SECTION A

SECTION B – Short-answer questions (30 marks)

Answer **all** questions in the spaces provided. Show all working in calculation questions. Give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states, for example $H_2(g)$, NaCl(s).

Question 1 (7 marks)

Before battery torches, many miners and cavers used carbide lamps. Water drips slowly from an upper chamber onto lumps of calcium carbide, CaC_2, in a lower chamber. The ethyne gas, C_2H_2 (also called acetylene), that forms flows to a jet, where it burns with a bright flame.

$$CaC_2(s) + 2H_2O(l) \rightarrow Ca(OH)_2(s) + C_2H_2(g)$$

A caver loads her lamp with 40.0 g of calcium carbide.

a) Calculate the amount, in mol, of CaC_2 in 40.0 g.

1 mark

b) Calculate the volume of ethyne, measured at SLC, that 40.0 g of CaC_2 can produce.

2 marks

c) For a long trip, the caver carries extra carbide. She expects to burn 30.0 L of ethyne, measured inside the cave at 14.0 °C and 99.0 kPa. Calculate the mass of CaC_2 needed to produce this volume of ethyne.

2 marks

d) The caver controls the size of the flame with a valve that sets how fast water drips onto the carbide. Explain, in terms of limiting and excess reactants, why slowing the drip reduces the amount of ethyne produced each minute, and so makes the flame smaller.

2 marks

Question 2 (9 marks)

Pharmacies sell hydrogen peroxide solution, $H_2O_2(aq)$, as an antiseptic. Anika's bottle is labelled "3% (m/v)". Hydrogen peroxide decomposes quickly when a small amount of solid manganese dioxide, MnO_2, is added as a catalyst. Water and one gas are the only products.

Anika carried out two experiments. Each used 40.0 mL of the antiseptic and the same mass of MnO_2.

Experiment 1: The open conical flask stood on an electronic balance. When the bubbling stopped, the mass of the flask and its contents had decreased by 0.566 g.

Experiment 2: The flask was sealed with a stopper and delivery tube. The volume of gas produced was 438 mL, measured at 21.0 °C and 98.5 kPa. Assume that the collected gas is dry.

a) Write a balanced equation for the decomposition of hydrogen peroxide, including states.

2 marks

b) Use the results of both experiments to calculate the molar mass of the gas.

3 marks

c) Use the amount of gas from part b) to calculate the concentration of hydrogen peroxide in the antiseptic, in % (m/v).

2 marks

d) In Experiment 1, fine droplets of the solution sprayed out of the open flask as the gas bubbled out. Explain how this would affect the molar mass calculated in part b).

2 marks

Question 3 (7 marks)

In a fume cupboard, a teacher demonstrates the reaction between copper, Cu, and concentrated nitric acid, HNO_3. A brown, toxic gas, nitrogen dioxide, NO_2, is produced and the solution turns blue as copper(II) nitrate, $Cu(NO_3)_2$, forms.

$$Cu(s) + 4HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2NO_2(g) + 2H_2O(l)$$

The teacher adds 1.27 g of copper turnings to 5.00 mL of 12.0 M $HNO_3(aq)$, and the reaction continues until it stops.

a) Calculate the amount, in mol, of copper used.

1 mark

b) Show by calculation which reactant is the limiting reactant.

2 marks

c) Calculate the volume of NO_2, measured at SLC, produced when the reaction stops.

2 marks

d) Calculate the extra volume of 12.0 M $HNO_3(aq)$, in mL, that would be needed to react with all of the copper.

2 marks

Question 4 (7 marks)

Before baking powder was common, bakers used “baker’s ammonia”, ammonium hydrogen carbonate, NH_4HCO_3, to make thin, crisp biscuits rise. In a hot oven it decomposes completely into three gases, so no solid is left in the biscuit.

$$NH_4HCO_3(s) \rightarrow NH_3(g) + CO_2(g) + H_2O(g)$$

A batch of biscuit dough contains 3.16 g of NH_4HCO_3. The oven temperature is 180 °C and the pressure is 100 kPa.

a) Calculate the amount, in mol, of NH_4HCO_3 in the dough.

1 mark

b) Calculate the total volume of gas produced in the oven, at 180 °C and 100 kPa.

3 marks

c) The gases leave the oven and cool to SLC. Assume that the gases do not react with each other and that none of them dissolves in any liquid that forms. Calculate the volume of gas present at SLC.

2 marks

d) Calculate the mass of ammonia, NH_3, released from the dough.

1 mark

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST

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SECTION A – MULTIPLE-CHOICE ANSWER SHEET

Student name: _____ Teacher: _____

Shade or circle ONE letter for each question. Each question is worth 1 mark. No marks are deducted for incorrect answers.

Question	A	B	C	D
1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)

Section A score: _____ / 10